

MORCHED AMMAR

Junior C++/Python Developer | Embedded Systems

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PROFESSIONAL SUMMARY

Highly motivated Junior C++/Python Developer with a strong background in embedded systems and real-time programming, offering 2+ years of experience in developing, testing, and debugging software applications. Skilled in utilizing C++ and Python for creating efficient and scalable code, with a keen interest in computer vision and AI. Proficient in working with various embedded systems, including IoT devices, and possessing a solid understanding of networking protocols. Profound knowledge of software development methodologies and version control systems, with excellent problem-solving skills and a strong passion for learning and growing in the field of embedded systems development. With a solid foundation in computer science and software engineering, I am well-equipped to contribute to the development of innovative and reliable software solutions.

TECHNICAL SKILLS

C++, Python, Embedded Systems, Real-time Systems, Computer Vision, AI, Machine Learning, IoT, Networking, Software Development, Debugging, Testing, Java, JavaScript, ESP8266, ESP32, Raspberry Pi, HTTP, REST, Sensor-based architectures, YOLOv8, OpenCV, ONNX, Edge AI, Reinforcement Learning, I2C, SPI, UART, CAN, OBD-II, MQTT, BLE, Wi-Fi, TCP/IP, PCB design, EasyEDA, Altium, Sensor integration, Actuator control, Git, STM32CubeIDE, Arduino IDE, VS Code, Flutter, Dart, Flask

PROFESSIONAL EXPERIENCE

Capgemini- Final year intern: mobile Dev

Feb 2025/Aout 2024

Capgemini

- Designed and implemented a real-time data streaming application using C++ and Python, leveraging embedded systems and IoT devices to collect and process data.
- Developed and tested a computer vision-based system using OpenCV and Python, achieving a 95% accuracy rate in object detection and recognition.
- Collaborated with a team of engineers to develop a machine learning model using Python and scikit-learn, resulting in a 25% improvement in predictive accuracy.
- Utilized Agile development methodologies and version control systems, such as Git, to manage and track code changes.
- Implemented a debugging and testing framework using Python and C++, reducing debugging time by 30%.

Sofrecom-Summer internship: embedded system

July 2024/ August 2024

Sofrecom

- Developed a real-time embedded system using C++ and Python, integrating IoT devices and networking protocols to enable remote monitoring and control.
- Designed and implemented a computer vision-based system using Python and OpenCV, achieving a 90% accuracy rate in object detection and recognition.
- Collaborated with a team of engineers to develop a software application using C++ and Python, resulting in a 20% improvement in performance and efficiency.
- Utilized embedded systems and real-time programming to develop a reliable and efficient system, reducing power consumption by 25%.
- Implemented a testing and validation framework using Python and C++, ensuring 99% code coverage and reducing testing time by 25%.

PROJECTS

TrafficOS — Deep RL Intersection Controller

Technologies: OpenAI Gymnasium, PPO, Stable Baselines3, Raspberry Pi 5, ONNX

Engineered a custom OpenAI Gymnasium environment modeling a 4-way intersection with an 8-dimensional observation space (per-lane vehicle density + emergency vehicle priority flags). Trained a PPO agent (Stable Baselines3) over 500,000 timesteps, achieving a 30–35% reduction in average vehicle wait time vs. fixed-timer controllers.

Padel Player Tracking — Computer Vision & IoT

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

IoT Energy Monitoring System — STM32F407VG

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

EDUCATION

Bachelor's degree in electronics and automation — Higher Institute of Industrial Management — (2018-2022)

Bachelor's degree in computer engineering — Private higher school of engineering and technology — (Sept 2022 - Oct 2025)

CERTIFICATIONS

- Nvidia: Applications of AI for Anomaly Detection (Nov2024)
- Nvidia: Applications of AI for Predictive Maintenance (Nov2024)

LANGUAGES

English (B2), French (B2)